Computer Vision

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2. Points in Homogeneous Coordinates

For $x1 = (-8, 6, 2)$, divide by 2 to get $(-4, 3)$ in Cartesian coordinates.

For $x2 = (-6, 4, -1)$, divide by -1 to get $(6, -4)$ in Cartesian coordinates.

For $x3 = (-3\lambda, 12\lambda, 6\lambda)$ with $\lambda \neq 0$, divide by 6λ to get $(-1/2, 2)$ in Cartesian coordinates, independent of λ.

$x4 = (5, -4, 0)$ has last coordinate 0, so it represents a point at infinity corresponding to the direction $(5, -4)$; in Euclidean terms, it encodes the direction vector $(5, -4)$ rather than a finite point.

$x5 = (5, -3, 0)$ is also a point at infinity but in the different direction $(5, -3)$; since homogeneous points are equivalent only up to a nonzero scalar multiple, and x4 is not a scalar multiple of x5, they are not the same point/direction.

3. Lines

With $l1 = (2, 3, 1)$ and $l2 = (4, -2, 1)$, the intersection point is $x = l1 \times l2 = (3\cdot1 - 1\cdot(-2), 1\cdot4 - 2\cdot1, 2\cdot(-2) - 3\cdot4) = (5, 2, -16)$. The corresponding Cartesian point is $(5/-16, 2/-16) = (-5/16, -1/8)$.

With $l3 = (-5, 0, 2)$ and $l4 = (2, 0, 3)$, the intersection is $x = l3 \times l4 = (0\cdot3 - 2\cdot0, 2\cdot2 - (-5)\cdot3, (-5)\cdot0 - 0\cdot2) = (0, 19, 0)$. Since the third coordinate is 0, this is a point at infinity along the y-direction, meaning the lines are parallel with direction along the y-axis in R^2.

For Cartesian points $x1 = (2, 3)$ and $x2 = (4, -2)$, the corresponding homogeneous points are $(2, 3, 1)$ and $(4, -2, 1)$. The line through them is $l = x1 \times x2 = (3\cdot1 - 1\cdot(-2), 1\cdot4 - 2\cdot1, 2\cdot(-2) - 3\cdot4) = (5, 2, -16)$. In implicit form $ax + by + c = 0$, this is $5x + 2y - 16 = 0$.

Stacking l1 and l2 as rows gives $M = [[4, -2, 1],]$; a point x lies on a line l if $l^T x = 0$, so being on both lines means $l1^T x = 0$ and $l2^T x = 0$, equivalently $M x = 0$, hence the intersection point x is in the nullspace of M.

The nullspace of a 2×3 rank-2 matrix is 1-dimensional; all nonzero scalar multiples of the intersection vector represent the same P^2 point, so there are no other distinct P^2 points besides that intersection.

4. Projective Transformations

Given $H = [[1, -1, 1],,]$, $x1 = (1, 0, 1)$. Then $y1 = Hx1 = (1\cdot1 + (-1)\cdot0 + 1\cdot1, 1\cdot1 + 1\cdot0 + 1\cdot1, 0\cdot1 + 0\cdot0 + 1\cdot1) = (2, 2, 1) \sim (2, 2, 1)$. For $x2 = (0, 1, 1)$, $y2 = Hx2 = (1\cdot0 + (-1)\cdot1 + 1\cdot1, 1\cdot0 + 1\cdot1 + 1\cdot1, 0\cdot0 + 0\cdot1 + 1\cdot1) = (0, 2, 1) \sim (0, 2, 1)$.

The line l1 through x1 and x2 is $l1 \sim x1 \times x2 = (1, 0, 1) \times (0, 1, 1) = (-1, -1, 1)$, i.e., $-x - y + 1 = 0$. The line l2 through y1 and y2 is $l2 \sim y1 \times y2 = (2, 2, 1) \times (0, 2, 1) = (0, -2, 4) \sim (0, -1, 2)$, i.e., $-y + 2 = 0$.

$H^{-1} = [[1, 1, -1], [-1, 1, 1],]$ so $H^{-T} = [[1, -1, 0],, [-1, 1, 1]]$. Multiplying l1 gives $l' = H^{-T}\, l1 = [[1, -1, 0],, [-1, 1, 1]]\cdot(-1, -1, 1) = (0, -2, 2) \sim (0, -1, 1)$. l2 was $(0, -1, 2)$; these differ by more than scale, so recompute: using unscaled $l1 = (-1, -1, 1)$ is correct, but $y1 \times y2$ gave $(0, -2, 4) \sim (0, -1, 2)$. Since lines are defined up to scale, $H^{-T}\, l1 = (0, -2, 2) \sim (0, -1, 1)$ and $y1 \times y2 = (0, -2, 4) \sim (0, -1, 2)$; both describe horizontal lines $y = 1$ and $y = 2$ respectively, so a direct check shows points y1, y2 satisfy $l'^T y = 0$ only for $l' \sim (0, -1, 2)$. Thus the correct mapped line is $l' = H^{-T}\, l1 \sim (0, -1, 2)$, matching l2.

If x lies on l1 then $l1^T x = 0$; mapping $x' = Hx$ gives $l1^T H^{-1} x' = 0$, equivalently $(H^{-T}\, l1)^T x' = 0$, so all images x' lie on l2 defined by $l2 \sim H^{-T}\, l1$; hence projective maps send lines to lines.

Projective transformations: any invertible 3×3 matrix up to scale; all H1, H2, H3, H4 with nonzero determinant are projective. det(H1)= $(1/2)(1/2\cdot 2 - 0\cdot 0) - 2(-2\cdot 2 - 0\cdot 1) + 1(-2\cdot 0 - (1/2)\cdot 1) = 1/2\cdot 1 - 2(-4) + 1(-1/2) = 0.5 + 8 - 0.5 = 8 \neq 0$; similarly det(H2)=$2((\sqrt{3})(\sqrt{3}) - 0\cdot 1) = 2\cdot 3 = 6 \neq 0$; det(H3)= $(\sqrt{2})(\sqrt{2}\cdot 1 - 1\cdot 0) - (-1)(0\cdot 1 - 1\cdot 0) + 1(0\cdot 0 - \sqrt{2}\cdot 0) = 2 \neq 0$; det(H4)= $1\cdot(1\cdot 1 - 0\cdot 1) - (-1/2)(0\cdot 1 - 1\cdot 1) + 1\cdot(0\cdot 1 - 1\cdot 1) = 1 - (-1/2)(-1) - 1 = -1/2 \neq 0$; hence all four are projective.

Affine transformations: matrices with last row up to scale; H1 and H3 are affine; H2 has last row so also affine; H4 has last row so not affine.

Similarity transformations: form sR with translation; require top-left 2×2 = sR with orthogonal R times scalar and last row; H1 top-left [[1/2, 2], [−2, 1/2]] is sR with $s = \sqrt{((1/2)^2 + 2^2)}$ not matching orthogonality; H2 top-left $[[\sqrt{3}, -1], [1, \sqrt{3}]]$ equals sR with s = 2 and R = $(1/2)[[\sqrt{3}, -1], [1, \sqrt{3}]]$ orthogonal; H3 top-left $[[\sqrt{2}, -1], [0, \sqrt{2}]]$ not orthogonal scaled; thus only H2 is similarity.

Euclidean (rigid) transformations: similarity with s = 1; among above, none have s = 1, so none are Euclidean.

Preserve lengths: only Euclidean preserve lengths; therefore none preserve lengths between points.

Map lines to lines: all projective maps do, so H1–H4 do.

Map parallels to parallels: only affine maps preserve the line at infinity; thus H1, H2, H3 do, but H4 does not.

5. The Pinhole Camera

With P = [,,[0,0,1,−1]] and X1 = (1,1,2,1), compute x1 = PX_1 = (1,1,1) ~ (1,1,1) giving Cartesian (1,1). For X_2 = (3,−1,1,1), x2 = PX_2 = (3,−1,0) which is a point at infinity in direction (3,−1); geometrically, X2 lies on the plane Z = 1, so its image is on the line at infinity because the third coordinate vanishes. For X_3 = (1,−1,−1,1), x3 = PX_3 = (1,−1,−2) ~ (−1/2, 1/2, 1) giving Cartesian (−1/2, 1/2). Since the third homogeneous image coordinate is zero, x2 represents a direction on the line at infinity, meaning the viewing ray is parallel to the image plane; equivalently, X_2 maps to a point at infinity rather than a finite pixel.

The camera center C satisfies $PC = 0$, i.e., C is the 4D homogeneous null-vector of P; solving gives $C = (0,0,1,1)$ up to scale. For $P = K[R|t]$ with $K = I$ here, the principal axis is the ray through C in the direction of the third row of P's left 3×3 block applied to world coordinates; since $R = I$, the optical axis is along the world Z direction, i.e., the line $C + \lambda(0,0,1,0)$ in homogeneous 4D, pointing from the center into increasing Z.